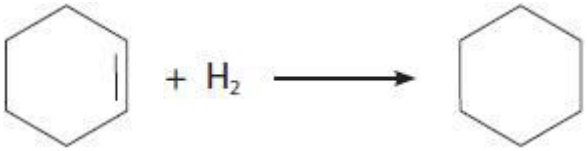
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Questions

Q1.

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

The enthalpy change of hydrogenation of cyclohexene is −120 kJ mol^{-1}.



What is the approximate enthalpy change of hydrogenation of benzene, in kJ mol^{-1}?

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- ☐ **A** −150
- ☐ **B** −210
- ☐ **C** −360
- ☐ **D** −510

(Total for question = 1 mark)

Q2.

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

	Orbitals overlapping	Type of bond
☐ **A**	p	π
☐ **B**	p	σ
☐ **C**	s and p	π
☐ **D**	s and p	σ

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(Total for question = 1 mark)

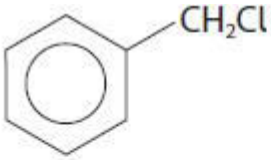
Q3.

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

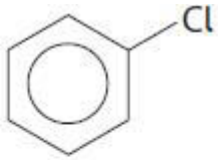
Chlorine reacts with methylbenzene in the presence of ultraviolet light.

Which of these is a product of this reaction?

☐ **A**

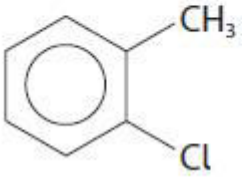


☐ **B**

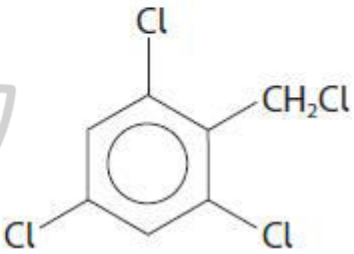


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☐ **C**



☐ **D**



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(Total for question = 1 mark)

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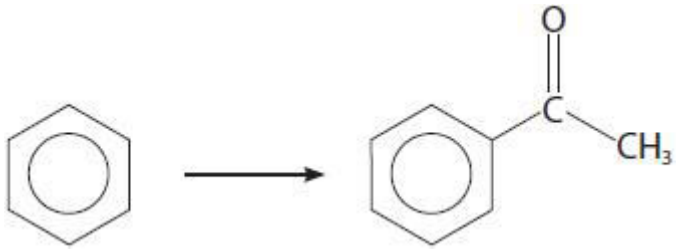
Q4.

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

Which reagent and catalyst will convert benzene into phenylethanone?

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	Reagent	Catalyst
☒ A	ethanal	aluminium chloride
☒ B	ethanal	platinum
☒ C	ethanoyl chloride	aluminium chloride
☒ D	ethanoyl chloride	platinum

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(Total for question = 1 mark)

Q5.

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

Some standard electrode potentials are shown.

Electrode system	$E^{\ominus}$ / V
$Bk^{3+} + e^- \rightleftharpoons Bk^{2+}$	−2.80
$Cu^{2+} + e^- \rightleftharpoons Cu^{+}$	+0.15
$Bk^{4+} + e^- \rightleftharpoons Bk^{3+}$	+1.67
$Au^{+} + e^- \rightleftharpoons Au$	+1.69
$Au^{2+} + e^- \rightleftharpoons Au^{+}$	+1.80
$Ag^{3+} + e^- \rightleftharpoons Ag^{2+}$	+1.80
$Ag^{2+} + e^- \rightleftharpoons Ag^{+}$	+1.98
$Cu^{3+} + e^- \rightleftharpoons Cu^{2+}$	+2.40

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Which of these disproportionation reactions is thermodynamically feasible under standard conditions?

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☒ **A** $2Bk^{3+} \rightarrow Bk^{2+} + Bk^{4+}$
☒ **B** $2Cu^{2+} \rightarrow Cu^{+} + Cu^{3+}$
☒ **C** $2Au^{+} \rightarrow Au + Au^{2+}$
☒ **D** $2Ag^{2+} \rightarrow Ag^{+} + Ag^{3+}$

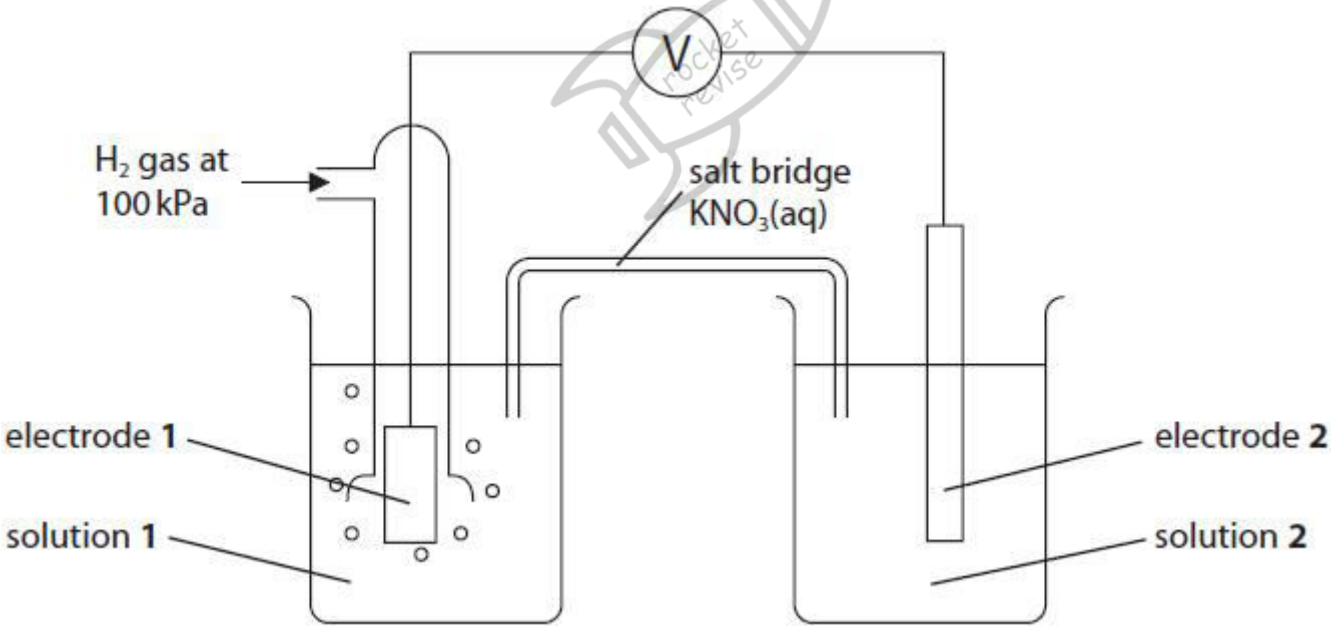
(Total for question = 1 mark)

Q6.

Answer the questions with a cross in the boxes you think are correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

The apparatus shown was used to measure the standard electrode potential for the reduction of $Cr_2O_7^{2-}$ ions to Cr^{3+} ions in acid solution:

$$Cr_2O_7^{2-}(aq) + 14H^+(aq) + 6e^- \rightarrow 2Cr^{3+}(aq) + 7H_2O(l)$$



(a) Which material should be used for each electrode?

(1)

		Electrode 1	Electrode 2
☐	**A**	$Na_2Cr_2O_7$	Cr_2O_3
☐	**B**	H_2	Cr
☐	**C**	Pt	Cr
☐	**D**	Pt	Pt

(b) Solution **1** is

(1)

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- ☐ **A** 0.33 mol dm^{-3} $H_3PO_4(aq)$
- ☐ **B** 0.50 mol dm^{-3} $H_2SO_4(aq)$
- ☐ **C** 1.00 mol dm^{-3} $HCl(aq)$
- ☐ **D** 1.00 mol dm^{-3} $CH_3COOH(aq)$

(c) Solution **2** contains 14.71 g of $K_2Cr_2O_7$.

What mass of $Cr_2(SO_4)_3 \cdot 18H_2O$ should also be used?
[M_r values: $K_2Cr_2O_7$ = 294.2 $Cr_2(SO_4)_3 \cdot 18H_2O$ = 716.3]

(1)

- ☐ **A** 8.95 g
- ☐ **B** 17.91 g
- ☐ **C** 19.62 g
- ☐ **D** 35.82 g

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(d) Solution **2** is **best** acidified with

(1)

- ☐ **A** H_2SO_4
- ☐ **B** HCl
- ☐ **C** HBr
- ☐ **D** H_2CrO_4

(Total for question = 4 marks)

Q7.

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

When an alkene is added to a solution of potassium manganate(VII), the purple solution turns colourless.

In terms of electron transfer and oxidation number, how does the manganese change in this reaction?

	Electron transfer	Oxidation number
☐ **A**	gains electrons	increases
☐ **B**	gains electrons	decreases
☐ **C**	loses electrons	increases
☐ **D**	loses electrons	decreases

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(Total for question = 1 mark)

Q8.

Answer the question with a cross in the box you think is correct ☒ . If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒ .

This question is about the reaction

$$2Fe^{3+}(aq) + Ti(s) \rightarrow 2Fe^{2+}(aq) + Ti^{2+}(aq) \qquad E^{\ominus}_{cell} = +2.40V$$

(a) The electrode potential for the Fe^{3+}/ Fe^{2+} electrode system is +0.77 V.

What is the electrode potential for the Ti^{2+}/Ti electrode system?

(1)

- ☐ **A** −3.17 V
- ☐ **B** −1.63 V
- ☐ **C** +1.63 V
- ☐ **D** +3.17 V

(b) What metals should be used for the electrodes in the cell for this reaction?

(1)

	Metals used for the electrode	
	Fe^{3+}/ Fe^{2+} electrode system	Ti^{2+}/ Ti electrode system
☐ **A**	iron	titanium
☐ **B**	iron	platinum
☐ **C**	platinum	titanium
☐ **D**	platinum	platinum

(c) The half-cell for the Fe^{3+}/ Fe^{2+} electrode system is prepared by mixing **equal** volumes of solutions of iron(II) sulfate, $FeSO_4$, and iron(III) sulfate, $Fe_2(SO_4)_3$.

What concentrations of the **original** solutions are needed for the resulting mixture to be standard?

(1)

	Concentration of the original solution	
	$FeSO_4$	$Fe_2(SO_4)_3$
☐ **A**	1 mol dm^{-3}	0.5 mol dm^{-3}
☐ **B**	1 mol dm^{-3}	1 mol dm^{-3}
☐ **C**	2 mol dm^{-3}	1 mol dm^{-3}
☐ **D**	2 mol dm^{-3}	2 mol dm^{-3}

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(Total for question = 3 marks)

Q9.

This question is about manganese compounds. Some data are given below.

	Electrode reaction	$E^{\ominus}$ / V
1	$MnO_4^- + e^- \rightleftharpoons MnO_4^{2-}$	+0.56
2	$MnO_4^{2-} + 2H_2O + 2e^- \rightleftharpoons MnO_2 + 4OH^-$	+0.59
3	$Fe^{3+} + e^- \rightleftharpoons Fe^{2+}$	+0.77
4	$MnO_2 + 4H^+ + 2e^- \rightleftharpoons Mn^{2+} + 2H_2O$	+1.23
5	$MnO_4^- + 8H^+ + 5e^- \rightleftharpoons Mn^{2+} + 4H_2O$	+1.51
6	$MnO_4^- + 4H^+ + 3e^- \rightleftharpoons MnO_2 + 2H_2O$	+1.70
7	$MnO_4^{2-} + 4H^+ + 2e^- \rightleftharpoons MnO_2 + 2H_2O$	+2.26

(a) (i) Write the ionic equation for the disproportionation of manganate(VI) ions, MnO_4^{2-}, in **acidic** conditions, using relevant half-equations from the table.
State symbols are not required.

(2)

(ii) Calculate $E^{\ominus}_{cell}$ for the disproportionation of manganate(VI) ions in **acidic** conditions, stating whether or not the reaction is thermodynamically feasible.

(2)

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(iii) Using the standard electrode potentials in the table, assess the thermodynamic feasibility of preparing manganate(VI) by reacting manganate(VII) and manganese(IV) oxide in **alkaline** conditions.

(4)

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(b) Steel is an alloy of iron and carbon. A group of students determined the iron content of a sample of steel wire by a titration method.

A known mass of the wire was dissolved in dilute sulfuric acid and the resulting solution made up to 250.0 cm^3 with more dilute sulfuric acid and mixed thoroughly.

$$Fe + H_2SO_4 \rightarrow FeSO_4 + H_2$$

25.0 cm^3 samples of the resulting solution were titrated with 0.0195 mol dm^{-3} potassium manganate(VII) solution.

(i) State the colour change at the end-point of the titration.

(1)

.....

(ii) One student used 1.53 g of the wire (weighed directly on the balance pan) and obtained a mean titre of 27.35 cm^3.

Using half-equations 3 and 5 from the table, calculate the percentage of iron in the steel wire. Give your answer to **three** significant figures.

(5)

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(iii) A second student carried out the same experiment but used distilled water to make up the solution in the volumetric flask.

A brown suspension formed during the titration.

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Explain how, if at all, the titre value would be affected by this student's error.

(3)

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(c) The uncertainties of the apparatus used in the experiment in (b) are shown.

Apparatus	Value measured	Uncertainty on each reading	Percentage uncertainty on value measured / %
Balance	1.53 g	±0.005 g	0.65
Burette	27.35 cm^3	±0.05 cm^3	
Pipette	25.0 cm^3	±0.06 cm^3	
Volumetric flask	250.0 cm^3	±0.3 cm^3	

(i) Complete the table.

(2)

(ii) A third student obtained a value of 95.863% for the proportion of iron in the wire. State whether or not this student has given their answer to an appropriate number of significant figures. Justify your answer in terms of the **total** percentage uncertainty of the experiment.

(2)

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(Total for question = 21 marks)

Q10.

Answer the question with a cross in the box you think is correct ☒ . If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒ .

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An electrochemical cell is set up to measure $E^{\ominus}_{cell}$ for the reaction

$$Fe(s) + Sn^{2+}(aq) \rightarrow Fe^{2+}(aq) + Sn(s)$$

(a) What is the cell diagram for this cell?

(1)

- ☒ **A** $Fe(s) \mid Fe^{2+}(aq) \parallel Sn(s) \mid Sn^{2+}(aq)$
- ☒ **B** $Fe^{2+}(aq) \mid Fe(s) \parallel Sn^{2+}(aq) \mid Sn(s)$
- ☒ **C** $Fe(s) \mid Fe^{2+}(aq) \parallel Sn^{2+}(aq) \mid Sn(s)$
- ☒ **D** $Fe^{2+}(aq) \mid Fe(s) \parallel Sn(s) \mid Sn^{2+}(aq)$

(b) The standard electrode potential for the Fe / Fe^{2+} electrode system is −0.44 V and $E^{\ominus}_{cell}$ for the reaction is +0.30 V.

What is the standard electrode potential for the Sn / Sn^{2+} electrode system?

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(1)

- ☒ **A** −0.74 V
- ☒ **B** −0.14 V
- ☒ **C** +0.14 V
- ☒ **D** +0.74 V

(Total for question = 2 marks)

Q11.

Answer the question with a cross in the box you think is correct ☒ . If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒ .

The standard hydrogen electrode uses an electrode of platinum coated in a finely divided form of the metal called platinum black.

What is the purpose of this coating?

- ☒ **A** to increase the rate of the equilibrium between the hydrogen gas and the hydrogen ions
- ☒ **B** to provide an inert protective coating for the electrode
- ☒ **C** to increase the electrical conductivity of the electrode
- ☒ **D** to ensure that the conditions remain standard

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(Total for question = 1 mark)

Q12.

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Answer the question with a cross in the box you think is correct ☒ . If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒ .

Hydrogen-oxygen fuel cells can operate in acidic or alkaline conditions.

What is the reaction at the anode in an alkaline hydrogen-oxygen fuel cell?

☐ **A** $O_2(g) + 2H_2O(l) + 4e^- \rightarrow 4OH^-(aq)$

☐ **B** $4OH^-(aq) \rightarrow O_2(g) + 2H_2O(l) + 4e^-$

☐ **C** $2H_2O(l) + 2e^- \rightarrow H_2(g) + 2OH^-(aq)$

☐ **D** $H_2(g) + 2OH^-(aq) \rightarrow 2H_2O(l) + 2e^-$

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(Total for question = 1 mark)

Q13.

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

Which is **true** of a hydrogen-oxygen fuel cell?

☐ **A** the cathode has a more positive potential than the anode
☐ **B** hydrogen is oxidised at the cathode
☐ **C** oxygen is reduced at the negative electrode
☐ **D** the cell potential is different when operating under alkaline or acidic conditions

(Total for question = 1 mark)

Q14.

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Answer the question with a cross in the box you think is correct ☒ . If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒ .

In which of the following pairs does the metal have **different** oxidation numbers?

☐ **A** CrO_4^{2-} and $Cr_2O_7^{2-}$

☐ **B** CrO_4^{2-} and CrO_3Cl^-

☐ **C** V_2O_5 and VO_4^{3-}

☐ **D** VO_2^+ and VO^{2+}

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(Total for question = 1 mark)

Q15.

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

The equation for a redox reaction is

$$5Fe^{2+}(aq) + MnO_4^-(aq) + 8H^+(aq) \rightarrow 5Fe^{3+}(aq) + Mn^{2+}(aq) + 4H_2O(l)$$

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Which is the correct cell diagram to measure $E^{\ominus}_{cell}$ cell for this reaction?

☐ **A** $Fe \mid Fe^{2+}, Fe^{3+} \parallel [MnO_4^- + 8H^+], [Mn^{2+} + 4H_2O] \mid Mn$

☐ **B** $Fe \mid Fe^{2+}, Fe^{3+} \parallel [Mn^{2+} + 4H_2O], [MnO_4^- + 8H^+] \mid Mn$

☐ **C** $Pt \mid Fe^{2+}, Fe^{3+} \parallel [MnO_4^- + 8H^+], [Mn^{2+} + 4H_2O] \mid Pt$

☐ **D** $Pt \mid Fe^{2+}, Fe^{3+} \parallel [Mn^{2+} + 4H_2O], [MnO_4^- + 8H^+] \mid Pt$

(Total for question = 1 mark)

Q16.

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

When copper is added to concentrated nitric acid, a brown gas is given off and the final solution is blue.

In terms of oxidation number and electron transfer, how does the **nitrogen** change in this reaction?

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	Oxidation number	Electron transfer
☐ **A**	decreases	gains electrons
☐ **B**	decreases	loses electrons
☐ **C**	increases	gains electrons
☐ **D**	increases	loses electrons

(Total for question = 1 mark)

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Q17.

Answer the questions with a cross in the boxes you think are correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

An electrochemical cell is set up using the electrode systems shown.

$$Cr_2O_7^{2-} + 14H^+ + 6e^- \rightleftharpoons 2Cr^{3+} + 7H_2O$$

$$TiO^{2+} + 2H^+ + e^- \rightleftharpoons Ti^{3+} + H_2O$$

(a) What materials will be used for the electrodes in this cell?

(1)

		$Cr_2O_7^{2-}, Cr^{3+}$	TiO^{2+}, Ti^{3+}
☒	**A**	chromium	titanium
☒	**B**	chromium	platinum
☒	**C**	platinum	titanium
☒	**D**	platinum	platinum

(b) The reaction between $Cr_2O_7^{2-}$ ions and Ti^{3+} ions has $E^{\ominus}_{cell} = +1.14\,V$.
The standard electrode potential for the $Cr_2O_7^{2-}, Cr^{3+}$ electrode system is +1.33 V.

What is the standard electrode potential for the TiO^{2+}, Ti^{3+} electrode system?

(1)

☒ **A** −2.47 V
☒ **B** −0.19 V
☒ **C** +0.19 V
☒ **D** +2.47 V

(Total for question = 2 marks)

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Q18.

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For each question, select one answer from A to D and put a cross in the box ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

A hydrogen-oxygen fuel cell is used to provide electrical energy for an electric motor in a car.

(a) The electrolyte in the fuel cell is acidic. What is the half-equation at the anode?

(1)

- ☐ **A** $\frac{1}{2}O_2(g) + 2H^+(aq) + 2e^- \rightarrow H_2O(l)$
- ☐ **B** $H_2O(l) \rightarrow \frac{1}{2}O_2(g) + 2H^+(aq) + 2e^-$
- ☐ **C** $H_2(g) \rightarrow 2H^+(aq) + 2e^-$
- ☐ **D** $2H^+(aq) + 2e^- \rightarrow H_2(g)$

(b) Hydrogen-oxygen fuel cells have advantages over methanol-oxygen fuel cells in vehicles.

Which of these is an advantage of the hydrogen-oxygen fuel cell?

(1)

- ☐ **A** more energy is released per mole of fuel used
- ☐ **B** emissions do not contribute to climate change
- ☐ **C** hydrogen is easier to store than methanol
- ☐ **D** only hydrogen can be obtained from renewable resources

(Total for question = 2 marks)

Q19.

For each question, select one answer from A to D and put a cross in the box ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

An electrochemical cell is made from the electrode systems shown by these half-equations.

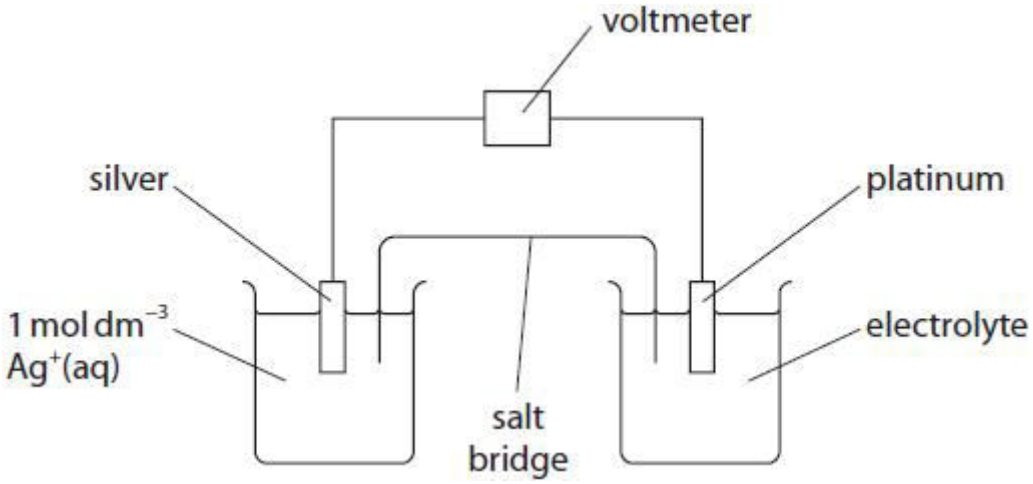
$$Ag^+(aq) + e^- \rightleftharpoons Ag(s) \qquad E^\ominus = +0.80V$$

$$VO_2^+(aq) + 2H^+(aq) + e^- \rightleftharpoons VO^{2+}(aq) + H_2O(l) \qquad E^\ominus = +1.00V$$

The apparatus used to measure the value for $E^\ominus_{cell}$ under standard conditions is shown.

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(a) Which silver compound could be used as the electrolyte in the left-hand half-cell?

(1)

- ☒ **A** silver nitrate
- ☒ **B** silver hydroxide
- ☒ **C** silver chloride
- ☒ **D** silver carbonate

(b) The electrolyte in the right-hand half-cell is prepared using equal volumes of

(1)

- ☐ **A** 1 mol dm^{-3} acidified VO_2^+(aq) and 1 mol dm^{-3} acidified VO^{2+}(aq)
- ☐ **B** 2 mol dm^{-3} acidified VO_2^+(aq) and 2 mol dm^{-3} acidified VO^{2+}(aq)
- ☐ **C** 1 mol dm^{-3} VO_2^+(aq) and 1 mol dm^{-3} HCl(aq)
- ☐ **D** 1 mol dm^{-3} VO^{2+}(aq) and 1 mol dm^{-3} HCl(aq)

(c) Which is the equation for the overall cell reaction under standard conditions?

(1)

- ☐ **A** $VO^{2+}(aq) + Ag^+(aq) + H_2O(l) \rightarrow VO_2^+(aq) + Ag(s) + 2H^+(aq)$
- ☐ **B** $VO_2^+(aq) + Ag(s) + 2H^+(aq) \rightarrow VO^{2+}(aq) + Ag^+(aq) + H_2O(l)$
- ☐ **C** $VO^{2+}(aq) + 3Ag(s) + 2H^+(aq) \rightarrow VO_2^+(aq) + 3Ag^+(aq) + H_2O(l)$
- ☐ **D** $VO_2^+(aq) + 3Ag^+(aq) + H_2O(l) \rightarrow VO^{2+}(aq) + 3Ag(s) + 2H^+(aq)$

(d) Which is the value of $E^{\ominus}_{cell}$ in volts?

(1)

- ☒ **A** −1.80
- ☒ **B** −0.20
- ☒ **C** +0.20
- ☒ **D** +1.80

(e) Which is the cell diagram for this cell, using the conventional representation of half-cells?

(1)

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☐ **A** $Ag(s) \mid Ag^+(aq) \mid\mid [VO_2^+(aq) + 2H^+(aq)] \mid [VO^{2+}(aq) + H_2O(l)] \mid Pt(s)$

☐ **B** $Ag(s) \mid Ag^+(aq) \mid\mid [VO_2^+(aq) + 2H^+(aq)], [VO^{2+}(aq) + H_2O(l)] \mid Pt(s)$

☐ **C** $Ag(s) \mid Ag^+(aq) \mid\mid [VO^{2+}(aq) + H_2O(l)] \mid [VO_2^+(aq) + 2H^+(aq)] \mid Pt(s)$

☐ **D** $Ag(s) \mid Ag^+(aq) \mid\mid [VO^{2+}(aq) + H_2O(l)], [VO_2^+(aq) + 2H^+(aq)] \mid Pt(s)$

(Total for question = 5 marks)

Q20.

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For each question, select one answer from A to D and put a cross in the box ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

A group of students carry out an experiment to find the concentration of chlorine, $Cl_2(aq)$, in a solution.

Excess potassium iodide solution is added to a 10.0 cm^3 sample of the chlorine solution.

$$Cl_2(aq) + 2I^-(aq) \rightarrow 2Cl^-(aq) + I_2(aq)$$

The iodine produced is titrated with a solution of thiosulfate ions of known concentration, using starch indicator.

$$2S_2O_3^{2-}(aq) + I_2(aq) \rightarrow S_4O_6^{2-}(aq) + 2I^-(aq)$$

The concentration of the $Cl_2(aq)$ is between 0.038 and 0.042 mol dm^{-3} .

(a) What concentration of thiosulfate ions, in mol dm^{-3} , is required to give a titre of approximately 20 cm^3 ?

(1)

☐ **A** 0.010
☐ **B** 0.020
☐ **C** 0.040
☐ **D** 0.080

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(b) What is the most suitable volume of 0.1 mol dm^{-3} potassium iodide solution, in cm^3 , to add to the 10.0 cm^3 of chlorine solution?

(1)

☐ **A** 7.6
☐ **B** 8.0
☐ **C** 8.4
☐ **D** 10.0

(c) What is the colour change at the end-point of the titration?

(1)

☐ **A** colourless to pale yellow
☐ **B** pale yellow to colourless

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☒ **C** colourless to blue-black
☒ **D** blue-black to colourless

(Total for question = 3 marks)

Q21.

The hydride of arsenic, arsine, is a toxic gas used in the production of semiconductors.

(a) Draw a dot-and-cross diagram for arsine, AsH_3 .

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(1)

(b) Arsine is a reducing agent and reacts with cerium(IV) sulfate solution, forming arsenic.

The data from an experiment are shown.
Volume of arsine gas = 350 cm^3 at 115 000 Pa and 20 °C
Volume of cerium(IV) sulfate solution = 488 cm^3
Concentration of cerium(IV) sulfate solution = 0.102 mol dm^{-3}

(i) Complete the half-equation.

(1)

$$AsH_3 \rightarrow As$$

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(ii) Calculate the final oxidation state of the cerium ion formed in the reaction.

(6)

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(Total for question = 8 marks)

Q22.

Answer the questions with a cross in the boxes you think are correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

This question is about alkaline hydrogen-oxygen fuel cells.

(a) What is the half-equation at the **negative** electrode?

- ☐ **A** $H_2(g) + 2OH^-(aq) \rightarrow 2H_2O(l) + 2e^-$
- ☐ **B** $2H_2O(l) + 2e^- \rightarrow H_2(g) + 2OH^-(aq)$
- ☐ **C** $O_2(g) + 2H_2O(l) + 4e^- \rightarrow 4OH-(aq)$
- ☐ **D** $4OH^-(aq) \rightarrow O_2(g) + 2H_2O(l) + 4e^-$

(b) Which statement is correct for an alkaline hydrogen-oxygen fuel cell when compared with an acidic hydrogen-oxygen fuel cell?

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(1)

- ☐ **A** $E^{\ominus}_{cell}$ is greater
- ☐ **B** $\Delta S^{\ominus}_{total}$ is greater
- ☐ **C** the catalyst is more efficient
- ☐ **D** K_c is greater

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(Total for question = 2 marks)

Q23.

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

In a chemical reaction, $E^{\ominus}_{cell}$ is **directly** proportional to

- ☐ **A** K_c
- ☐ **B** $\Delta_r H^{\ominus}$
- ☐ **C** $\Delta S^{\ominus}_{system}$
- ☐ **D** $\Delta S^{\ominus}_{total}$

(Total for question = 1 mark)

Q24.

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

The equation for a redox reaction is shown.

$$2V^{2+}(aq) + Cu^{2+}(aq) \rightarrow 2V^{3+}(aq) + Cu(s)$$

What is the cell diagram for this reaction?

- ☐ **A** V(s) | V^{3+}(aq), V^{2+}(aq) | | Cu(s) | Cu^{2+}(aq)
- ☐ **B** V(s) | V^{2+}(aq), V^{3+}(aq) | | Cu^{2+}(aq) | Cu(s)
- ☐ **C** Pt(s) | V^{3+}(aq), V^{2+}(aq) | | Cu(s) | Cu^{2+}(aq)
- ☐ **D** Pt(s) | V^{2+}(aq), V^{3+}(aq) | | Cu^{2+}(aq) | Cu(s)

(Total for question = 1 mark)

Q25.

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Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

An electrochemical cell is set up:

- a half-cell is made from a piece of zinc and a solution of zinc chloride, $ZnCl_2$
- a second half-cell is made from a piece of metal **G** and a solution of its chloride, **G**Cl_2
- the two half-cells are connected and a current allowed to pass for some time.

The zinc electrode increased in mass by 1.635 g.

The electrode of metal **G** decreased in mass by 0.6075 g.

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What is metal **G**?

- ☒ **A** copper
- ☒ **B** iron
- ☒ **C** magnesium
- ☒ **D** manganese

(Total for question = 1 mark)

Use this space for any rough working. Anything you write in this space will gain no credit.

Q26.

The amount of calcium ethanedioate, CaC_2O_4, present in a sample of spinach is determined by redox titration.

11.4 g of spinach leaves are stirred in 50.0 cm^3 of a warm acidified solution of 0.0100 mol dm^{-3} potassium manganate(VII), $KMnO_4$, oxidising all the ethanedioate ions, $C_2O_4^{2-}$.

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$$16H^+ + 2MnO_4^- + 5C_2O_4^{2-} \rightarrow 2Mn^{2+} + 10CO_2 + 8H_2O$$

The excess manganate(VII) ions, MnO_4^-, are then titrated with an acidified solution of 0.0500 mol dm^{-3} iron(II) sulfate, $FeSO_4$.

$$5Fe^{2+} + 8H^+ + MnO_4^- \rightarrow 5Fe^{3+} + Mn^{2+} + 4H_2O$$

25.95 cm^3 of iron(II) sulfate solution was needed for complete reaction.

Calculate the percentage by mass of CaC_2O_4 present in the spinach leaves. Give your answer to an appropriate number of significant figures.

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(Total for question = 6 marks)

Q27.

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

What is the minimum volume, in cm^3, of oxygen needed for the complete combustion of a mixture of 50 cm^3 of propene and 50 cm^3 of but-1-ene gases?

All gas volumes are measured at the same temperature and pressure.

- ☐ **A** 225
- ☐ **B** 300
- ☐ **C** 525
- ☐ **D** 700

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(Total for question = 1 mark)

Q28.

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

For the reaction shown, $E^{\ominus}_{cell} = +0.89V.$

$$4V^{3+}(aq) + O_2(g) + 2H_2O(l) \rightarrow 4VO^{2+}(aq) + 4H^+(aq)$$

Which statement can be deduced from this information only?

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- **A** the reaction will **not** occur under any conditions
- **B** the reactants are kinetically stable with respect to the products
- **C** the reactants are thermodynamically unstable with respect to the products
- **D** an aqueous solution of V^{3+} will oxidise rapidly on standing in air

(Total for question = 1 mark)

Q29.

This question is about electrochemical cells and redox reactions.

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(a) Draw a labelled diagram of the apparatus you would use to measure the standard electrode potential of a $Cu^{2+}(aq)|Cu(s)$ electrode with a standard hydrogen electrode. Include essential conditions.

(5)

(b) The standard electrode potentials for two half-cells are shown.

$$Cr_2O_7^{2-}(aq) + 14H^+(aq) + 6e^- \rightleftharpoons 2Cr^{3+}(aq) + 7H_2O(l) \qquad E^\ominus = +1.33V$$

$$Cl_2(g) + 2e^- \rightleftharpoons 2Cl^-(aq) \qquad E^\ominus = +1.36V$$

(i) Explain, in terms of electrode potentials, why acidified dichromate(VI) ions react with **concentrated** hydrochloric acid to form chlorine, even though $E^\ominus_{cell} = -0.03V$.

(3)

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(ii) Write the cell diagram, using the conventional representation of half-cells, for the reaction to produce chlorine.

(2)

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(c) When 25.0 cm^3 of a 0.100 mol dm^{-3} solution of X_2O_5 reacts with a reducing agent, X is reduced to a lower oxidation state.

To oxidise X back to its original oxidation state required 50.0 cm^3 of 0.0200 mol dm^{-3} acidified potassium manganate(VII) solution.
The half-equation for acidified manganate(VII) is

$$MnO_4^- + 8H^+ + 5e^- \rightarrow Mn^{2+} + 4H_2O$$

Calculate the oxidation state of X after it has been reduced.

(3)

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(Total for question = 13 marks)

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Q30.

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

The half-equation for the reaction taking place at the positive electrode in an **alkaline** hydrogen-oxygen fuel cell is

$$O_2 + 2H_2O + 4e^- \rightarrow 4OH^-$$

What is the half-equation for the reaction taking place at the negative electrode?

- ☒ **A** $H_2 + 2OH^- \rightarrow 2H_2O + 2e^-$
- ☒ **B** $H_2 + OH^- \rightarrow H_2O + H^+ + 2e^-$
- ☒ **C** $H_2 \rightarrow 2H^+ + 2e^-$
- ☒ **D** $H_2 + H_2O \rightarrow 3H^+ + OH^- + 2e^-$

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(Total for question = 1 mark)

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Q31.

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

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The cell diagram for an electrochemical cell is shown.

$$Pt(s) \mid Fe^{2+}(aq), Fe^{3+}(aq) \parallel Br_2(aq), 2Br^-(aq) \mid Pt(s) \qquad E^{\ominus}_{cell} = +0.32\,V$$

What are the half-equations for the reactions that occur in the half-cells?

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	Half-equation in left-hand half-cell	Half-equation in right-hand half-cell
☐ **A**	$Fe^{2+}(aq) \rightleftharpoons Fe^{3+}(aq) + e^-$	$2Br^-(aq) \rightleftharpoons Br_2(aq) + 2e^-$
☐ **B**	$Fe^{2+}(aq) \rightleftharpoons Fe^{3+}(aq) + e^-$	$Br_2(aq) + 2e^- \rightleftharpoons 2Br^-(aq)$
☐ **C**	$Fe^{3+}(aq) + e^- \rightleftharpoons Fe^{2+}(aq)$	$2Br^-(aq) \rightleftharpoons Br_2(aq) + 2e^-$
☐ **D**	$Fe^{3+}(aq) + e^- \rightleftharpoons Fe^{2+}(aq)$	$Br_2(aq) + 2e^- \rightleftharpoons 2Br^-(aq)$

(Total for question = 1 mark)

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Q32.

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

Persulfate ions react slowly with iodide ions at room temperature unless a catalyst of Fe^{2+} or Fe^{3+} is added to the reaction mixture.

$$S_2O_8^{2-}(aq) + 2I^-(aq) \rightarrow 2SO_4^{2-}(aq) + I_2(aq)$$

Which of these is the most likely reason why this reaction is slow without a catalyst?

☐ **A** $E^{\ominus}_{cell}$ is positive

☐ **B** $\Delta_r H$ is positive

☐ **C** ΔS_{total} is positive

☐ **D** activation energy is high

(Total for question = 1 mark)

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Q33.

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

Which reaction occurs at the **negative** electrode in a hydrogen-oxygen fuel cell?

☐ **A** $H_2 + 2OH^- \rightarrow 2H_2O + 2e^-$

☐ **B** $2H^+ + 2e^- \rightarrow H_2$

☐ **C** $O_2 + 4H^+ + 4e^- \rightarrow 2H_2O$

☐ **D** $4OH^- \rightarrow O_2 + 2H_2O + 4e^-$

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(Total for question = 1 mark)

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Q34.

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

The emf, $E^{\ominus}_{cell}$, of a cell is +0.57 V.

The numerical values of the standard electrode potentials of the two half-cells joined in this cell are 0.17 V and 0.40 V.

What are the signs of the standard electrode potentials of the right-hand half-cell and the left-hand half-cell?

	Sign of standard electrode potential	
	in left-hand half-cell	in right-hand half-cell
☐ A	negative	negative
☐ B	negative	positive
☐ C	positive	negative
☐ D	positive	positive

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(Total for question = 1 mark)

Q35.

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

Standard electrode potentials can also be given an alternative name. The electrochemical series lists

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standard electrode potentials in order.

Which of these is correct?

	Alternative name for standard electrode potential	Order of standard electrode potentials in the electrochemical series
☐ **A**	standard reduction potential	most negative to most positive
☐ **B**	standard reduction potential	most positive to most negative
☐ **C**	standard cell potential	most negative to most positive
☐ **D**	standard cell potential	most positive to most negative

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(Total for question = 1 mark)

Q36.

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

$E^{\ominus}_{cell}$ is directly proportional to

- ☐ **A** $\Delta_r H$ and ln K
- ☐ **B** $\Delta_r H$ and ln RT
- ☐ **C** ΔS_{total} and ln K
- ☐ **D** ΔS_{total} and ln RT

(Total for question = 1 mark)

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Q37.

Answer the question with a cross in the box you think is correct ☒. If you change your mind

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about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

In which of these pairs does the transition metal have the same oxidation number?

- ☐ **A** CrO_2Cl_2 and $[Cr(NH_3)_4Cl_2]^+$
- ☐ **B** $[Cu(NH_3)_2]^+$ and $[CuCl_4]^{2-}$
- ☐ **C** Mn_2O_3 and MnO_2
- ☐ **D** VO_3^- and VO_2^+

(Total for question = 1 mark)

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Q38.

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

Some standard electrode potentials are shown.

Right-hand electrode system	$E^\ominus$ / V
$Mg^{2+} + 2e^- \rightleftharpoons Mg$	−2.37
$Ce^{3+} + 3e^- \rightleftharpoons Ce$	−2.33
$Mn^{2+} + 2e^- \rightleftharpoons Mn$	−1.19
$Mn^{3+} + e^- \rightleftharpoons Mn^{2+}$	+1.49
$Ce^{4+} + e^- \rightleftharpoons Ce^{3+}$	+1.70

Which reaction is thermodynamically feasible?

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- ☐ **A** $2Ce + 3Mg^{2+} \rightarrow 2Ce^{3+} + 3Mg$
- ☐ **B** $4Ce^{3+} \rightarrow Ce + 3Ce^{4+}$
- ☐ **C** $3Mn^{2+} \rightarrow Mn + 2Mn^{3+}$
- ☐ **D** $Mg + 2Ce^{4+} \rightarrow Mg^{2+} + 2Ce^{3+}$

(Total for question = 1 mark)

Q39.

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Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

A sample of air contains 405 parts per million (ppm) of carbon dioxide.

What is the approximate volume, in cm^3 , of carbon dioxide in 250 cm^3 of this air?

- ☐ **A** 0.001
- ☐ **B** 0.01
- ☐ **C** 0.1
- ☐ **D** 1

(Total for question = 1 mark)

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Q40.

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

The formula of hydrated copper(II) bromide is $CuBr_2 \cdot 4H_2O$.

What is the percentage, by mass, of water of crystallisation in this compound?

- ☐ **A** 6.10 %
- ☐ **B** 8.06 %
- ☐ **C** 24.4 %
- ☐ **D** 32.2 %

(Total for question = 1 mark)

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Q41.

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

A titre has an uncertainty of 0.32 %.

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The uncertainty of each burette reading is ±0.05 cm^3 .

What is the most likely value of the titre in cm^3 ?

- **A** 6.40
- **B** 15.60
- **C** 31.25
- **D** 32.00

(Total for question = 1 mark)

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Q42.

A hydrogen-oxygen fuel cell contains an alkaline electrolyte. The half-equation at the anode is:

$$H_2(g) + 2OH^-(aq) \rightarrow 2H_2O(l) + 2e^-$$

What is the half-equation at the cathode?

(1)

- **A** $\frac{1}{2}O_2(g) + H_2O(l) + 2e^- \rightarrow 2OH^-(aq)$
- **B** $O_2(g) + 2H^+(aq) + 2e^- \rightarrow H_2O_2(l)$
- **C** $\frac{1}{2}O_2(g) + 2H^+(aq) + 2e^- \rightarrow H_2O(l)$
- **D** $O_2(g) + H_2(g) + 2e^- \rightarrow 2OH^-(aq)$

(Total for question = 1 mark)

Q43.

A redox titration of iron(II) ions with potassium manganate(VII) is used to determine the amount of iron in iron tablets. The reaction is:

$$MnO_4^-(aq) + 8H^+(aq) + 5Fe^{2+}(aq) \rightarrow Mn^{2+}(aq) + 4H_2O(l) + 5Fe^{3+}(aq).$$

(a) Why is no indicator necessary in this redox titration?

(1)

- **A** an indicator would interfere with the redox reaction
- **B** no suitable indicator changes colour at the end point
- **C** the colour change of the iron(II) ions is sufficient
- **D** the colour change of the manganate(VII) ions is sufficient

(b) In one such titration, the following equipment was used.

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Equipment	Uncertainty for each reading
100 cm^3 measuring cylinder	$\pm 1\ cm^3$
250.0 cm^3 volumetric flask	$\pm 0.15\ cm^3$
25.0 cm^3 pipette	$\pm 0.06\ cm^3$
50.00 cm^3 burette	$\pm 0.05\ cm^3$

Which piece of equipment has the **lowest** measurement uncertainty for this experiment?

(1)

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- ☒ **A** the measuring cylinder to measure 100 cm^3 of sulfuric acid
- ☒ **B** the volumetric flask to make up the solution of the iron tablet
- ☒ **C** the pipette to measure out the iron(II) solution
- ☒ **D** the burette to add a titre volume of 25.00 cm^3

(c) A 25.0 cm^3 portion of an iron(II) tablet solution required 5.00×10^{-5} mol of manganate(VII) ions to react completely. What is the mass of iron, in grams, in the 25.0 cm^3?

(1)

- ☒ **A** 0.00058
- ☒ **B** 0.0028
- ☒ **C** 0.010
- ☒ **D** 0.014

(Total for question = 3 marks)

Q44.

Standard electrode potentials can be used to show whether or not a reaction is feasible.

(a) State the conditions required, in addition to 1 mol dm^{-3}, for obtaining standard electrode potentials.

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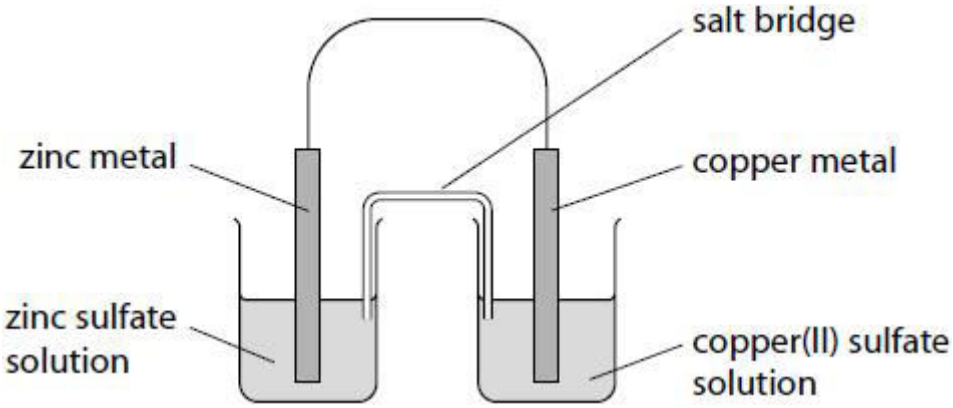
(1)

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(b) A Daniell cell is a combination of standard zinc and copper electrodes.

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The standard electrode potentials measured against a standard hydrogen electrode are shown in the table.

Right-hand electrode system	$E^{\ominus}$/V
$Zn^{2+}(aq) \mid Zn(s)$	−0.76
$Cu^{2+}(aq) \mid Cu(s)$	+0.34

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(i) Calculate the standard electrode potential of this cell.

(1)

(ii) Give **three** observations that would be made when current flows for several hours in the Daniell cell.

(2)

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(c) Some standard reduction potentials are:

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Electrode reaction	$E^{\ominus}$ / V
$Cu^{2+}(aq) + e^- \rightleftharpoons Cu^+(aq)$	+0.15
$Cu^+(aq) + e^- \rightleftharpoons Cu(s)$	+0.52
$Fe^{2+}(aq) + 2e^- \rightleftharpoons Fe(s)$	−0.44
$Fe^{3+}(aq) + e^- \rightleftharpoons Fe^{2+}(aq)$	+0.77

State and justify, in terms of $E^{\ominus}$ cell values, whether copper(I) ions and iron(II) ions will be disproportionate.

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Include any equations for reactions which occur.

(3)

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(d) State one reason why the feasibility determined from standard electrode potentials does not necessarily result in a reaction.

(1)

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(Total for question = 8 marks)

Q45.

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

What is the pressure of hydrogen gas used in the standard hydrogen electrode?

- ☒ **A** 1 Pa
- ☒ **B** 100 Pa
- ☒ **C** 1 000 Pa
- ☒ **D** 100 000 Pa

(Total for question = 1 mark)

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Q46.

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

The possibility of a reaction between potassium dichromate(VI) and hydrochloric acid may be assessed using standard electrode potentials but also depends on the activation energy, E_a, of the reaction.

$$Cr_2O_7^{2-} + 14H^+ + 6Cl^- \rightarrow 2Cr^{3+} + 7H_2O + 3Cl_2 \qquad E^{\ominus}_{cell} = +0.03V$$

When potassium dichromate(VI) and hydrochloric acid are mixed, very little chlorine is formed under standard conditions but a significant amount of chlorine is produced when concentrated hydrochloric acid is used.

What is the effect on E_{cell} and on E_a of using concentrated hydrochloric acid?

	E_{cell}	E_a
☐ A	less positive	decreased
☐ B	less positive	unchanged
☐ C	more positive	decreased
☐ D	more positive	unchanged

(Total for question = 1 mark)

Q47.

Answer the question with a cross in the box you think is correct (☒). If you change your mind about an answer, put a line through the box (☒) and then mark your new answer with a cross (☒).

Which is the diagram for the cell made up of the nickel and zinc half-cells that gives a positive value for $E^{\ominus}_{cell}$?

Use your Data Booklet.

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- ☐ **A** $Zn(s) \mid Zn^{2+}(aq) \parallel Ni^{2+}(aq) \mid Ni(s)$
- ☐ **B** $Zn^{2+}(aq) \mid Zn(s) \parallel Ni(s) \mid Ni^{2+}(aq)$
- ☐ **C** $Ni(s) \mid Ni^{2+}(aq) \parallel Zn^{2+}(aq) \mid Zn(s)$
- ☐ **D** $Ni^{2+}(aq) \mid Ni(s) \parallel Zn(s) \mid Zn^{2+}(aq)$

(Total for question = 1 mark)

Q48.

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Iron is generally prescribed when a person is suffering from anaemia. Many people find it easier to take medicine in liquid form rather than solid tablets.

An iron solution, advertised for this purpose, contains iron as Fe^{2+}.

An adult is advised to take 90.0 mg of iron daily.

A 25.0 cm^3 sample of the solution was made up to 100.0 cm^3 using sulfuric acid and deionised water.

25.0 cm^3 samples of the diluted iron(II) solution were titrated with a 0.0125 mol dm^{-3} potassium manganate(VII) solution.

The mean titre was 16.20 cm^3 .

The reaction is shown.

$$MnO_4^- + 8H^+ + 5Fe^{2+} \rightarrow Mn^{2+} + 5Fe^{3+} + 4H_2O$$

Calculate the volume of iron solution required to obtain a 90.0 mg dose of iron.

Give your answer to an appropriate number of significant figures.

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Volume required by an adult to obtain a 90.0 mg dose ..... cm^3

(Total for question = 6 marks)

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Q49.

This question is about electrochemical cells.

*(a) Describe how you would carry out an experiment to determine the standard electrode potential of the electrode system shown.

$$Co^{2+}(aq) + 2e^- \rightleftharpoons Co(s)$$

Assume that you have access to the equipment and chemicals that you need, describing their use in your answer.
You may include a labelled diagram.

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(6)

(b) Acidified potassium dichromate(VI) is an oxidising agent.

The acid commonly used is sulfuric acid.
Explain, using reference to the electrode potentials shown, why **concentrated** hydrochloric acid should **not** be used to acidify potassium dichromate(VI).

Electrode system	$E^{\ominus}$/V
$Cr_2O_7^{2-} + 14H^+ + 6e^- \rightleftharpoons 2Cr^{3+} + 7H_2O$	+1.33
$Cl_2 + 2e^- \rightleftharpoons 2Cl^-$	+1.36

(3)

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(c) The cell diagram for an electrochemical cell is shown.

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$$Ag(s) \mid Ag^{+}(aq) \parallel [VO_2^{+}(aq) + 2H^{+}(aq)], [VO^{2+}(aq) + H_2O(l)] \mid Pt(s)$$

Write the **reduction** half-equation for this electrochemical cell.
State symbols are not required.

(1)

(d) The Direct Methanol Fuel Cell (DMFC) uses methanol and oxygen to generate electricity. When the cell operates, the methanol reacts with water at one electrode to produce carbon dioxide and hydrogen ions.

At the other electrode the oxygen reacts with hydrogen ions to form water.

(i) Write the oxidation half-equation for this fuel cell.
State symbols are not required.

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(1)

(ii) Write the overall equation for this fuel cell.
State symbols are not required.

(1)

(iii) The reduction half-cell in the DMFC has a standard electrode potential $E^{\ominus} = +1.23\,V$.
When the solution concentration changes, the electrode potential changes so that E = +1.20 V.
The relationship between these values is given by the equation

$$E = E^{\ominus} + 4.277 \times 10^{-3} \ln \mathbf{X}$$

where **X** is the concentration of the solution in mol dm^{-3}.
Calculate the value of **X** in the non-standard half-cell.

(1)

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(Total for question = 13 marks)

Q50.

Answer the questions with a cross in the boxes you think are correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

The half-equations for a hydrogen-oxygen fuel cell in **alkaline** solution are shown.

$2H_2O(l) + 2e^- \rightarrow H_2(g) + 2OH^-(aq)$ $\quad E^{\ominus} = -0.83\ V$

$\frac{1}{2}O_2(g) + H_2O(l) + 2e^- \rightarrow 2OH^-(aq)$ $\quad E^{\ominus} = +0.40\ V$

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(a) The equation for the overall cell reaction is

(1)

- ☒ **A** $3H_2O(l) + \frac{1}{2}O_2(g) \rightarrow H_2(g) + 4OH^-(aq)$
- ☒ **B** $H_2(g) + 4OH^-(aq) \rightarrow 3H_2O(l) + \frac{1}{2}O_2(g)$
- ☒ **C** $H_2(g) + \frac{1}{2}O_2(g) \rightarrow H_2O(l)$
- ☒ **D** $H_2O(l) \rightarrow H_2(g) + \frac{1}{2}O_2(g)$

(b) Calculate $E^{\ominus}_{cell}$ for the reaction occurring in the hydrogen-oxygen fuel cell, under alkaline conditions.

(1)

- ☒ **A** –1.23 V
- ☒ **B** –0.43 V
- ☒ **C** +0.43 V
- ☒ **D** +1.23 V

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(Total for question = 2 marks)

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Q51.

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

Which sequence shows the ions in order of increasing strength as a reducing agent? Refer to your Data Booklet.

- ☐ **A** $V^{2+} < Fe^{2+} < Cr^{2+}$
- ☐ **B** $Cr^{2+} < Fe^{2+} < V^{2+}$
- ☐ **C** $Cr^{2+} < V^{2+} < Fe^{2+}$
- ☐ **D** $Fe^{2+} < V^{2+} < Cr^{2+}$

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(Total for question = 1 mark)

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Q52.

Answer the question with a cross in the box you think is correct (☒). If you change your mind about an answer, put a line through the box (☒) and then mark your new answer with a cross (☒)

An equation for the reaction of $FeSO_4$ with $KMnO_4$ is shown.

$$2KMnO_4 + 10FeSO_4 + 8H_2SO_4 \rightarrow K_2SO_4 + 2MnSO_4 + 5Fe_2(SO_4)_3 + 8H_2O$$

(a) Which shows the changes in the oxidation states for iron and manganese?

(1)

		Iron	Manganese
☐	**A**	II → III	VII → II
☐	**B**	II → III	VII → IV
☐	**C**	III → II	II → VII
☐	**D**	III → II	III → IV

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(b) A 25.00 cm^3 sample of $FeSO_4$ dissolved in acid required 22.50 cm^3 of 0.100 mol dm^{-3} $KMnO_4$ for complete reaction.

What is the concentration of the $FeSO_4$ solution?

(1)

☐ **A** 0.011 mol dm^{-3}
☐ **B** 0.100 mol dm^{-3}
☐ **C** 0.450 mol dm^{-3}
☐ **D** 0.900 mol dm^{-3}

(Total for question = 2 marks)

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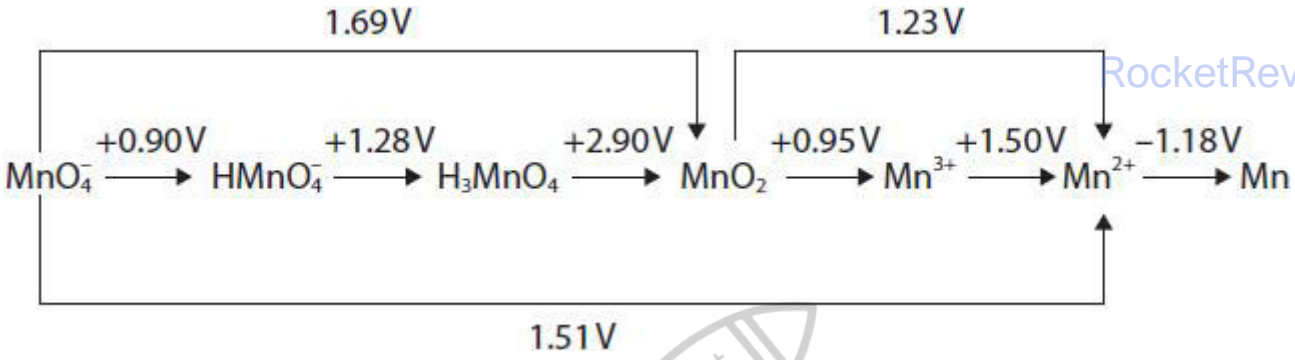
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Q53.

A Latimer diagram for a chemical element is a summary of the standard electrode potential data for that element.

In a Latimer diagram, the form of the element with the highest oxidation state is on the left, with successively lower oxidation states to the right.

A Latimer diagram for manganese at pH = 0 is shown.



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The diagram shows that the standard electrode potential for the reduction of MnO_4^- to MnO_2, in acidic conditions, is +1.69 V.

$$MnO_4^- + 4H^+ + 3e^- \rightleftharpoons MnO_2 + 2H_2O \qquad E^\ominus = +1.69\,V$$

(a) (i) Justify the assignment of the oxidation state of +5 to manganese in H_3MnO_4 using oxidation numbers.

(1)

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(ii) Write an equation for the reaction of H_3MnO_4 in acidic solution to give ions containing manganese(VI) and manganese(IV).

Use the Latimer diagram to obtain the formulae of the ions produced. State symbols are not required.

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(2)

(iii) Deduce whether or not this disproportionation reaction is thermodynamically feasible by calculating $E^\ominus$ for the reaction.

(2)

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(b) Before use in titration experiments, potassium manganate(VII) solutions must be standardised. One method uses ethanedioate ions to find the exact concentration of the manganate(VII) ions.

250.0 cm^3 of a standard solution contained 1.915 g of sodium ethanedioate, $Na_2C_2O_4$.
A potassium manganate(VII) solution of approximately 0.02 mol dm^{-3} was standardised using this solution.
Excess sulfuric acid was added to 25.0 cm^3 portions of the potassium manganate(VII) solution which were titrated with the sodium ethanedioate solution.
The mean titre was 22.95 cm^3.
The relevant ionic half–equations are shown.

$$MnO_4^- + 8H^+ + 5e^- \rightarrow Mn^{2+} + 4H_2O$$

$$C_2O_4^{2-} \rightarrow 2CO_2 + 2e^-$$

(i) State the colour change at the end-point of the titration.

(1)

.....

.....

(ii) Calculate the accurate concentration of the potassium manganate(VII), in mol dm^{-3}, giving your answer to an appropriate number of significant figures.

(4)

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(iii) A second titration carried out without the addition of sulfuric acid resulted in the formation of a brown suspension.
Explain how the value of the mean titre would be affected, if at all, by the reaction that forms this suspension.
Use the Data Booklet as a source of information.
There is no need to calculate E_{cell} values.

(3)

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(Total for question = 13 marks)

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Q54.

Answer the questions with a cross in the boxes you think are correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

Sodium thiosulfate can be used to determine the concentration of an iodine solution by titration using starch indicator.

(a) 5.00 g of sodium thiosulfate was dissolved in deionised water and the solution made up to 250.0 cm^3 in a volumetric flask.

The volumetric flask has an uncertainty of ± 0.25 cm^3.
What is the minimum uncertainty of the balance required to match the uncertainty of the volumetric flask?
Assume two weighings are needed.

(1)

- ☐ **A** ± 0.0025 g
- ☐ **B** ± 0.005 g
- ☐ **C** ± 0.01 g
- ☐ **D** ± 0.05 g

(b) The titration was carried out with sodium thiosulfate in the burette and starch was added just before the end-point.

What would be the colour of the solution in the conical flask at the end-point?

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(1)

- ☐ **A** blue-black
- ☐ **B** brown
- ☐ **C** colourless
- ☐ **D** yellow

(Total for question = 2 marks)

Q55.

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Answer the question with a cross in the box you think is correct (☒). If you change your mind about an answer, put a line through the box (☒) and then mark your new answer with a cross (☒).

The emf, $E^{\ominus}_{cell}$, of the cell made up of the copper and magnesium half-cells is measured under standard conditions.

The voltmeter used has a measurement uncertainty of 0.50%.

What is the **highest** emf that could be measured in this experiment?

Use your Data Booklet.

- ☐ **A** 2.70 V
- ☐ **B** 2.72 V
- ☐ **C** 2.74 V
- ☐ **D** 2.85 V

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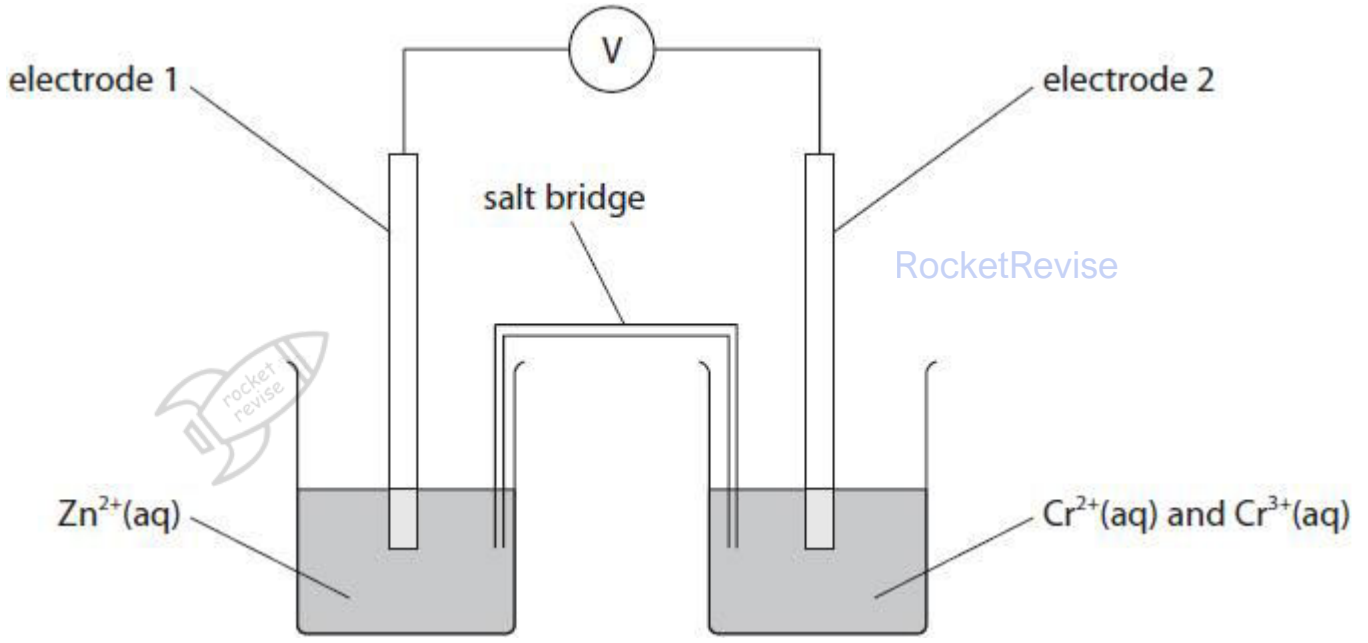
(Total for question = 1 mark)

Q56.

Answer the questions with a cross in the boxes you think are correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

The apparatus can be used to measure $E^{\ominus}_{cell}$ for the reaction shown.

$$2Cr^{3+}(aq) + Zn(s) \rightarrow 2Cr^{2+}(aq) + Zn^{2+}(aq)$$



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(a) Which electrodes are used for this cell?

(1)

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		electrode 1	electrode 2
☐	**A**	platinum	platinum
☐	**B**	platinum	chromium
☐	**C**	zinc	chromium
☐	**D**	zinc	platinum

(b) A student wishes to measure the standard cell potential, $E^{\ominus}_{cell}$, of this cell. The right-hand cell requires Cr^{3+} and Cr^{2+} ions.

What mass of $Cr_2(SO_4)_3 \cdot 18H_2O$ must be dissolved in 1.00 dm^3 of deionised water to give the concentration of Cr^{3+} ions required to measure this $E^{\ominus}_{cell}$?

(1)

☐ **A** 52.0 g

☐ **B** 196 g

☐ **C** 358 g

☐ **D** 716 g

(c) What can be deduced from the fact that, for this reaction, $E^{\ominus}_{cell}$ is positive?

(1)

☐ **A** ΔS_{total} and ln *K* are positive

☐ **B** ΔS_{total} and ln *K* are negative

☐ **C** ΔS_{total} is positive and ln *K* is negative

☐ **D** ΔS_{total} is negative and ln *K* is positive

(Total for question = 3 marks)

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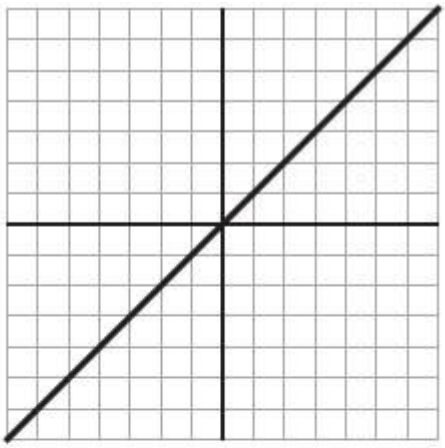
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Q57.

Answer the question with a cross in the box you think is correct (☒). If you change your mind about an answer, put a line through the box (☒) and then mark your new answer with a cross (☒).

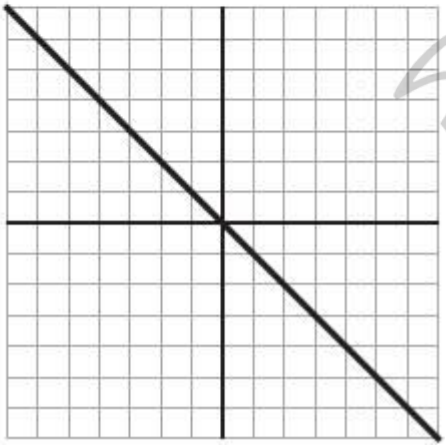
Which graph shows the relationship between $E^{\ominus}_{cell}$ plotted on the x-axis and ΔS_{total} on the y-axis for any electrochemical system?

☐ **A**



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☐ **B**



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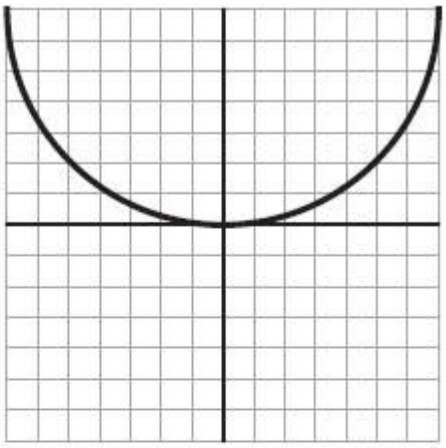
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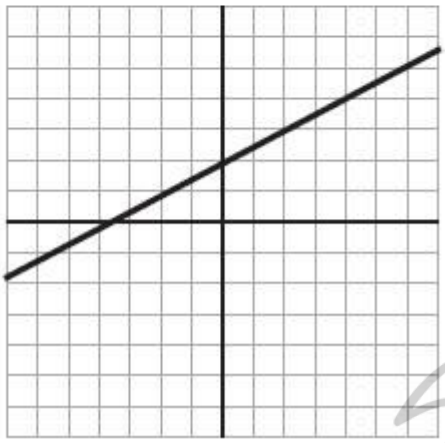
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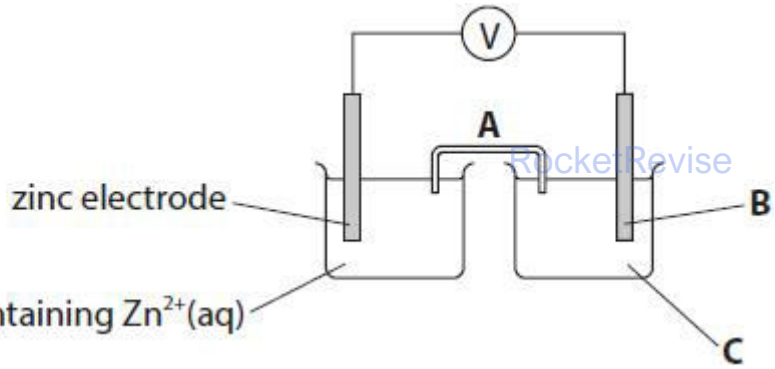


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(Total for question = 1 mark)

Q58.

The standard emf, $E^{\ominus}_{cell}$, of the cell shown was measured.



Electrode system	Standard electrode potential $E^{\ominus}$/V
$Zn^{2+}(aq) + 2e^- \rightleftharpoons Zn(s)$	−0.76
$Fe^{3+}(aq) + e^- \rightleftharpoons Fe^{2+}(aq)$	+0.77

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(a) (i) Identify, by name or formula, the substances needed in the salt bridge and the right-hand half-cell. Include concentrations where appropriate.

(4)

A Salt bridge containing a solution of the compound

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B Electrode made of

.....

C Solution containing the compounds

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(ii) State why a salt bridge is used rather than a wire.

(1)

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(b) Write an equation for the reaction that would take place if current was allowed to flow through the circuit. State symbols are not required.

(2)

(c) Under standard conditions, $E^{\ominus}_{cell}$ for this reaction is +1.53 V.

State what would happen to this value, if anything, when the Zn^{2+} solution in the half-cell is diluted. Justify your answer.

(2)

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(Total for question = 9 marks)

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Q59.

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

Which is correct about standard electrode potentials?

- ☐ **A** $E^{\ominus}_{cell}$ is proportional to $\ln K$ and $\ln \Delta S_{total}$
- ☐ **B** $E^{\ominus}_{cell}$ is proportional to $\ln K$ and ΔS_{total}
- ☐ **C** $\ln E^{\ominus}_{cell}$ is proportional to K and ΔS_{total}
- ☐ **D** $E^{\ominus}_{cell}$ is proportional to $\ln K$ and $\Delta S_{surroundings}$

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(Total for question = 1 mark)

Q60.

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒. and then mark your new answer with a cross ☒.

Which reaction is thermodynamically feasible under standard conditions?

Use your Data Booklet.

- ☐ **A** $Ag + V^{3+} \rightarrow Ag^{+} + V^{2+}$
- ☐ **B** $Cu^{+} + V^{3+} \rightarrow Cu^{2+} + V^{2+}$
- ☐ **C** $Fe^{2+} + V^{3+} \rightarrow Fe^{3+} + V^{2+}$
- ☐ **D** $Zn + 2V^{3+} \rightarrow Zn^{2+} + 2V^{2+}$

(Total for question = 1 mark)

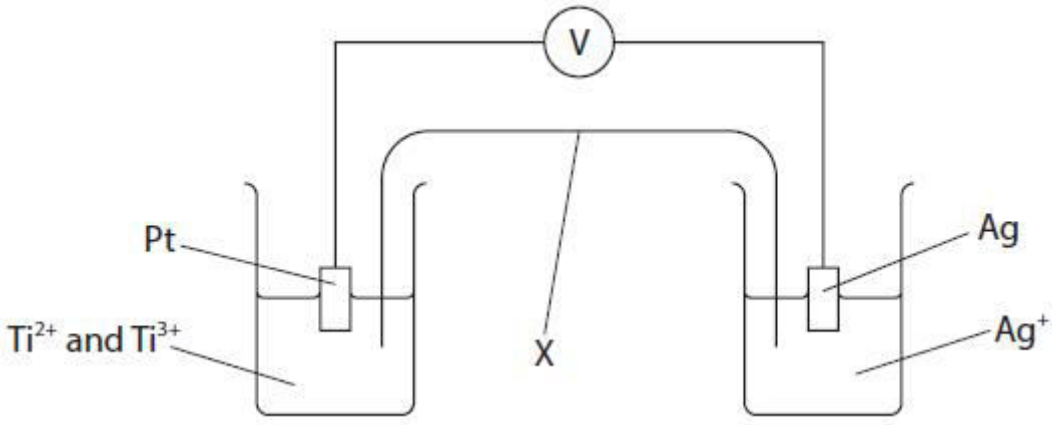
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Q61.

This question is about the electrochemical cell shown.

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(a) State the name and purpose of part **X**.

(2)

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(b) Draw a cell diagram for this system.

(2)

(c) The electrode system for the titanium ions is shown.

$Ti^{3+}(aq) + e^- \rightleftharpoons Ti^{2+}(aq) \quad E^{\ominus} = -0.37V$

Calculate the $E^{\ominus}_{cell}$ value. Use your Data Booklet.

(2)

(d) State the changes, if any, that would occur to the cell emf, the electrodes and the ion concentrations when current flows for several hours in this cell.

(3)

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(Total for question = 9 marks)

Q62.

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒. and then mark your new answer with a cross ☒.

In a redox titration, 0.0500 mol dm^{-3} potassium manganate(VII) is titrated against 25.0 cm^3 of a solution containing iron(II) ions.

$$MnO_4^- + 8H^+ + 5Fe^{2+} \rightarrow 5Fe^{3+} + Mn^{2+} + 4H_2O$$

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The mean titre of potassium manganate(VII) is 18.95 cm^3.

(a) What colour change is seen at the end-point?

(1)

- ☐ **A** pale pink to pale green
- ☐ **B** purple to pale pink
- ☐ **C** pale green to pale pink
- ☐ **D** pale pink to purple

(b) What is the concentration, in mol dm^{-3}, of the Fe^{2+} ions in the solution?

(1)

- ☐ **A** 1.90×10^{-1}
- ☐ **B** 7.58×10^{-3}
- ☐ **C** 4.74×10^{-3}
- ☐ **D** 1.90×10^{-4}

(c) What is the percentage error in a single titre of 18.95 cm^3?

(1)

[The uncertainty in a single burette reading is ±0.05 cm^3]

- ☐ **A** ±0.20%
- ☐ **B** ±0.26%
- ☐ **C** ±0.40%
- ☐ **D** ±0.53%

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(d) Which row shows the correct changes in oxidation states?

(1)

	manganese	iron
☐ **A**	II → VII	III → II
☐ **B**	II → VII	II → III
☐ **C**	VII → II	III → II
☐ **D**	VII → II	II → III

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(Total for question = 4 marks)

Q63.

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒. and then mark your new answer with a cross ☒.

Which reaction occurs at an electrode in a hydrogen-oxygen fuel cell?

- ☒ **A** $O_2(g) + 2H_2(g) \rightarrow 2H_2O(l)$
- ☒ **B** $O_2(g) + 2H_2O(l) + 4e^- \rightarrow 4OH^-(aq)$
- ☒ **C** $4H^+(aq) + 4OH^-(aq) \rightarrow 4H_2O(l)$
- ☒ **D** $H_2O(l) + 2OH^-(aq) \rightarrow 2H_2(g) + 1½O_2(g) + 2e^-$

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(Total for question = 1 mark)

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